

**DESIGN AND ANALYSIS OF A DUAL-BAND PODIUM-SHAPED MICROSTRIP
PATCH ANTENNA FOR n46 AND n78 BAND APPLICATIONS**

A PROJECT REPORT

of

ECB1323 – ANTENNA AND MILLIMETER WAVE COMMUNICATION

Submitted by

AARTHI S

MADHUMITHA T

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MEERA B

*in partial fulfilment for the award of the degree
of*

**BACHELOR OF ENGINEERING
in
ELECTRONICS AND COMMUNICATION ENGINEERING**



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
CHENNAI 600 025**

MAY 2026

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*in partial fulfilment for the award of the degree
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**BACHELOR OF ENGINEERING
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ELECTRONICS AND COMMUNICATION ENGINEERING

Under the Guidance of

Dr. G. KALPANADEV, M.E., Ph.D.,

Department of Electronics and Communication Engineering

**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)**

Under

**ANNA UNIVERSITY, CHENNAI
MAY 2026**





**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)**



**Under
ANNA UNIVERSITY, CHENNAI**

BONAFIDE CERTIFICATE

Certified that this project report titled “**DESIGN AND ANALYSIS OF A DUAL-BAND PODIUM-SHAPED MICROSTRIP PATCH ANTENNA FOR n46 AND n78 BAND APPLICATIONS**” is the bonafide work of **AARTHI S (8115U23EC001), MADHUMITHA T (8115U23EC059), MARIYAYEE L (8115U23EC061), MEERA B (8115U23EC062)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voce held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATES

**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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DEPARTMENT OF ECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will be competent enough to design, develop and integrate complex industrial products for modern electronics and communication systems.

PEO2: Graduates will be able to propose research oriented, industry relevant design solutions in state-of-the-art electronics and communication domains.

PEO3: Graduates will be able to engage in start-up and entrepreneurial activities developing innovative prototypes for the challenging needs of the society.

PROGRAM SPECIFIC OUTCOMES (PSO'S):

PSO1: Analyze and design complex electronic circuits, communication and embedded systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, communication systems and embedded VLSI.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and the World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Lifelong learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

CO-PO MAPPING FOR PROJECT WORK

After successful completion of this course, the students should be able to

CO1: Illustrate the basic principles and parameters of antennas and their role in communication systems.

CO2: Design and analyze the antenna arrays.

CO3: Comprehend millimeter-wave communication principles and the role of MIMO systems.

CO4: Identify the various emerging trends and applications of millimetre wave antennas.

CO5: Design, simulate and analyse mm Wave antennas for 5G communication using EM simulation tools.

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	✓	✓	✓	✓	-	-	-	✓	✓	-	✓	✓	✓	✓
CO2	✓	✓	✓	✓	-	✓	-	✓	✓	✓	-	✓	✓	✓
CO3	✓	✓	✓	✓	✓	-	-	✓	✓	-	-	✓	✓	✓
CO4	✓	✓	✓	✓	-	✓	-	✓	✓	✓	-	✓	✓	✓
CO5	✓	✓	✓	✓	✓	-	-	✓	✓	-	-	✓	✓	✓

ACKNOWLEDGEMENT

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We thank **Dr. M. MAHESWARI, M.Tech., Ph.D.**, Head of the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, for providing her encouragement in pursuing this project.

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ABSTRACT

This work presents the design and analysis of a dual-band podium-shaped Microstrip Patch Antenna operating at 3.5 GHz and 5.3 GHz for modern wireless communication applications. The 3.5 GHz band corresponds to Sub-6 GHz 5G (n78 band), supporting enhanced mobile broadband (eMBB), while the 5.3 GHz band is used for WLAN to enable high-speed Wi-Fi communication. The antenna is designed using an FR4 substrate and a microstrip line feeding technique to ensure proper impedance matching and ease of fabrication.

The proposed podium-shaped geometry modifies the surface current distribution by creating multiple effective current paths, which results in dual resonant frequencies within a compact structure. The antenna achieves good performance characteristics with return loss ($S_{11} < -10$ dB) and maintains $VSWR \leq 2$ at both operating frequencies, ensuring efficient power transmission. It also exhibits stable and nearly omnidirectional radiation patterns with moderate gain and efficiency. Simulation results confirm that the antenna provides reliable dual-band performance, making it suitable for integration in compact wireless devices. Overall, the proposed design offers a simple, cost-effective, and efficient solution for combining 5G and WLAN functionalities in a single antenna system.

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CHAPTER 1

INTRODUCTION

1.1. INTRODUCTION

Wireless communication has become an essential part of modern technology, enabling connectivity in applications such as 5G networks, WLAN (Wi-Fi), and smart electronic devices. Antennas play a crucial role in these systems by converting electrical signals into electromagnetic waves for transmission and reception. Among various antenna types, the Microstrip Patch Antenna is widely preferred due to its low profile, lightweight structure, compact size, low fabrication cost, and ease of integration with printed circuit boards. The performance of a microstrip antenna depends on parameters such as patch geometry, substrate material, and feeding technique, which influence characteristics like resonant frequency, impedance matching, bandwidth, and radiation pattern. However, conventional designs such as rectangular patch antennas generally support only single-band operation, making them less suitable for modern multi-band communication systems.

To overcome this limitation, a podium-shaped microstrip patch antenna is proposed to achieve dual-band operation at 3.5 GHz and 5.3 GHz. The 3.5 GHz frequency corresponds to the Sub-6 GHz 5G (n78 band) used for enhanced mobile broadband applications, while the 5.3 GHz band is used for WLAN communication, enabling high-speed Wi-Fi connectivity. The podium-shaped geometry modifies the surface current distribution by creating multiple effective current paths, which generate two distinct resonant frequencies within a compact structure. This design ensures good impedance matching ($S_{11} < -10$ dB), maintains $VSWR \leq 2$, and provides stable radiation characteristics, making it suitable for integrated 5G and WLAN wireless communication applications.

1.2. EXISTING WORK

In recent years, several research works have been carried out on Microstrip Patch Antenna designs for wireless communication applications such as Sub-6 GHz 5G (n78 band) and WLAN. Conventional microstrip patch antennas, especially rectangular and circular shapes, are widely used due to their simple structure and ease of fabrication. These antennas provide good performance in terms of radiation characteristics and impedance matching but are generally limited to single-band operation, which restricts their use in modern multi-band communication systems.

To overcome this limitation, various techniques such as slot loading, stacked patches, parasitic elements, and defected ground structures (DGS) have been proposed to achieve dual-band or multi-band performance. Although these methods improve bandwidth and enable multiple resonances, they often increase the complexity of the design, antenna size, and fabrication cost. Some designs also suffer from poor impedance matching, reduced efficiency, or unstable radiation patterns. Therefore, there is a need for a compact, simple, and efficient antenna design that can achieve dual-band operation without significantly increasing complexity. This limitation in existing designs has motivated the development of the proposed podium-shaped microstrip patch antenna, which aims to provide improved dual-band performance with a simple and compact structure.

1.3. DRAWBACKS FOUND IN EXISTING WORK

Conventional Microstrip Patch Antenna designs such as rectangular and circular shapes are widely used due to their simple structure, low cost, and ease of fabrication. However, when applied in modern wireless applications like Sub-6 GHz 5G (n78 band) and WLAN, these antennas exhibit several performance limitations. They generally support only single-band operation, making them unsuitable for systems that require dual-band functionality. In addition, these antennas often suffer from narrow bandwidth, poor impedance matching, and reduced efficiency, especially when designed for compact devices.

To achieve dual-band performance, existing methods such as slotting, stacking, and use of parasitic elements are employed, but these approaches increase the design complexity and fabrication cost. Moreover, maintaining resonance at multiple frequencies often requires modifications that increase antenna size or affect performance. Conventional designs also provide limited control over surface current distribution, resulting in lower gain and less stable radiation patterns. Furthermore, these antennas are sensitive to substrate properties and fabrication variations, which can degrade overall performance in practical implementations.

1.4. NEED FOR THE PROPOSED WORK

The rapid growth of advanced wireless communication systems such as Sub-6 GHz 5G (n78 band) and WLAN has created a significant demand for antennas capable of supporting multiple frequency bands within a compact structure. Conventional Microstrip Patch Antenna designs are limited by single-band operation, narrow bandwidth, and reduced efficiency, which restrict their application in modern multi-functional communication systems.

Existing approaches to achieve dual-band performance often involve complex structures such as slotting, stacking, or parasitic elements, which increase the design complexity, fabrication cost, and antenna size. In addition, these methods may affect impedance matching and radiation stability. Therefore, there is a need for a simple, compact, and efficient antenna design that can achieve reliable dual-band operation without compromising performance. The proposed podium-shaped microstrip patch antenna addresses these challenges by modifying the geometry to create multiple current paths, enabling operation at 3.5 GHz and 5.3 GHz with improved impedance matching, stable radiation characteristics, and compact size, making it suitable for integrated 5G and WLAN applications.

CHAPTER 2

LITERATURE SURVEY

2.1. LITERATURE SURVEY

1. Antenna Theory: Analysis and Design – 2016 (Balanis)

Key Findings:

- Developed analytical models for microstrip antenna design and analysis
- Provides equations for resonant frequency, impedance, and radiation

Disadvantages:

- Limited to simple geometries
- Less suitable for complex or modified shapes

2. Compact and Broadband Microstrip Antennas – 2002 (Wong)

Key Findings:

- Introduces methods for size reduction and bandwidth enhancement
- Uses techniques like slots and stacked patches

Disadvantages:

- Design becomes complex
- Fabrication difficulty increases

3. Microstrip Antennas – 1992 (Pozar)

Key Findings:

- Explains working principle and feeding techniques
- Provides basic design approach for microstrip antennas

Disadvantages:

- Focuses on single-band designs
- Limited discussion on dual-band operation

4. Microstrip Antenna Design Handbook – 2001 (Garg et al.)

Key Findings:

- Provides comprehensive design formulas and procedures
- Covers various substrates and feeding methods

Disadvantages:

- Mainly based on conventional designs
- Less focus on compact modern antennas

5. Handbook of Microstrip Antennas – 1989 (James & Hall)**Key Findings:**

- Covers fundamental concepts and radiation characteristics
- Provides early methods for antenna design

Disadvantages:

- Not suitable for modern wireless applications
- Limited support for multi-band designs

6. Dual-Frequency Patch Antennas – 1997 (Maci & Gentili)**Key Findings:**

- Demonstrates techniques for dual-band operation
- Uses patch modification methods

Disadvantages:

- Requires complex geometry changes
- Bandwidth is limited

7. Microstrip Patch Antennas for Wireless Applications – 2017 (Wong et al.)**Key Findings:**

- Focuses on antennas for modern wireless systems (5G/WLAN)
- Improves gain and bandwidth performance

Disadvantages:

- Antenna size may be larger
- Design optimization is complex

8. CST Microwave Studio User Manual – 2023**Key Findings:**

- Provides tools for accurate antenna simulation and analysis
- Supports evaluation of S11, VSWR, gain, and radiation pattern

Disadvantages:

- Results depend on user setup accuracy
- Requires high computational resources

9. Slot-Loaded Dual-Band Microstrip Antenna – 2020 (Kumar et al.)**Key Findings:**

- Achieves dual-band operation using slot technique
- Improves frequency tuning capability

Disadvantages:

- Increases design complexity
- Fabrication becomes difficult

10. Fractal-Based Multiband Microstrip Antenna – 2021 (Verma et al.)**Key Findings:**

- Achieves multiband operation using fractal geometry
- Improves bandwidth and frequency coverage

Disadvantages:

- Structure is complex
- Difficult to design and fabricate

CHAPTER 3

PROJECT DESCRIPTION

3.1. INTRODUCTION

The proposed project focuses on the design and simulation of a compact podium-shaped Microstrip Patch Antenna intended to operate at 3.5 GHz and 5.3 GHz for modern wireless communication applications. Microstrip antennas are widely preferred in portable and embedded systems due to their low profile, lightweight structure, compact size, and ease of integration with printed circuit boards. However, conventional patch geometries often face challenges in achieving dual-band operation, good impedance matching, and stable radiation characteristics while maintaining compact dimensions.

The antenna is designed using an FR-4 substrate and analyzed through electromagnetic simulation to evaluate key performance parameters such as resonant frequency, return loss, and voltage standing wave ratio (VSWR). The 3.5 GHz band corresponds to Sub-6 GHz 5G (n78 band), while the 5.3 GHz band is used for WLAN communication. The proposed podium-shaped geometry enhances current distribution to achieve dual resonance within a compact structure. The project aims to develop a simple, low-cost, and efficient antenna design that can be integrated into modern wireless devices supporting both 5G and Wi-Fi applications.

3.2. DESIGN OF THE ANTENNA

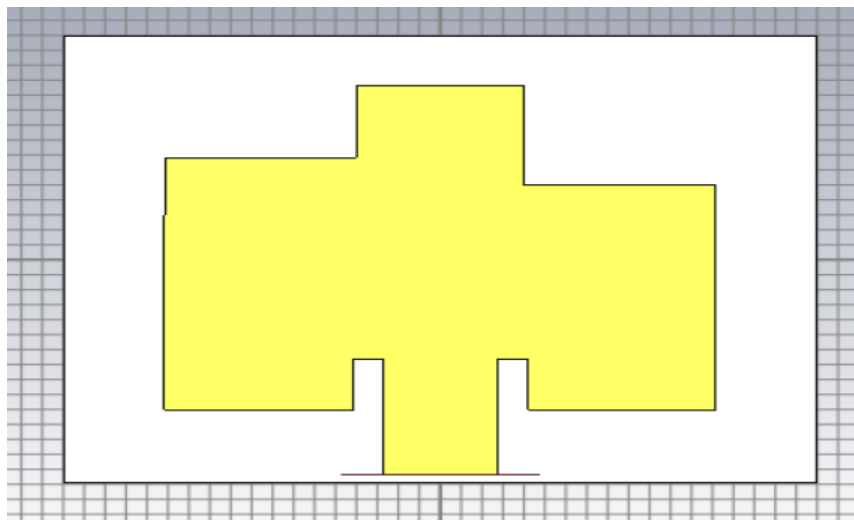


Fig.3.1 DESIGN OF A PODIUM-SHAPED MICROSTRIP PATCH ANTENNA FOR DUAL-BAND 3.5 GHz AND 5.3 GHz APPLICATIONS

3.3. DESIGN DIAGRAM WITH DIMENSIONS

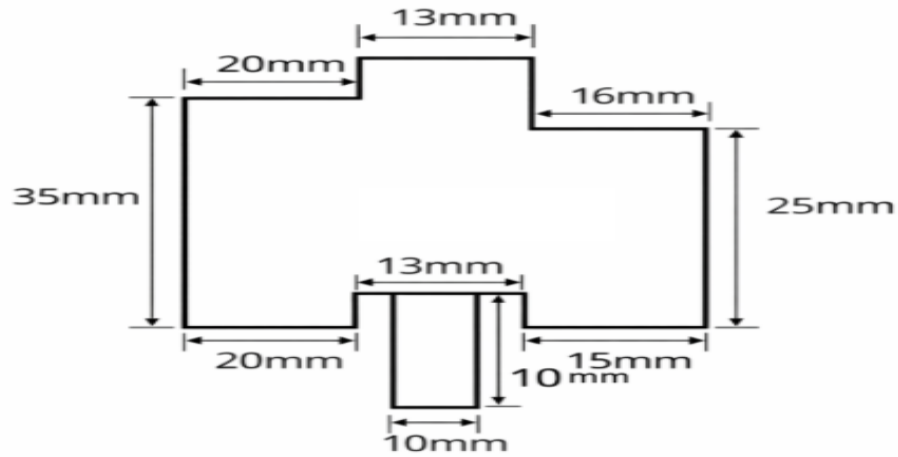


Fig.3.2. DESIGN DIAGRAM WITH DIMENSIONS

EXPLANATION:

The structure comprises three distinct geometric components integrated into a single unified profile:

1. **Stepped rectangular body:** The main structure consists of a series of stepped rectangular sections forming an irregular profile. The left vertical section has a height of 35 mm connected to a 20 mm base. The upper region includes horizontal steps of 20 mm and 13 mm, creating a reduced-width top section. On the right side, a vertical section of 25 mm is connected to a horizontal extension of 16 mm.
2. **Top stepped section:** The upper portion of the structure forms a narrow platform with a width of 13 mm. This stepped configuration introduces variation in geometry, which helps in defining the overall shape and structural layout.
3. **Central notch (cut-out):** A rectangular notch of dimensions 10 mm × 10 mm is provided at the bottom center, with an upper opening width of 13 mm. This feature alters the geometry, introduces a discontinuity, and can be used for fitting or functional purposes.

3.4. APPLICATIONS

The proposed podium-shaped microstrip patch antenna finds applications in modern wireless systems requiring compact size, dual-band operation, and stable radiation performance.

- **Sub-6 GHz 5G communication systems (3.5 GHz – n78 band)** for high data rate applications such as video streaming and mobile broadband services.
- **WLAN (5.3 GHz band)** used in high-speed Wi-Fi communication for indoor networking and internet access.

- **Integrated dual-band communication devices** enabling simultaneous support for both 5G and Wi-Fi within a single antenna structure.
- **Smart and IoT-based systems** including connected devices, automation systems, and wireless sensors.
- **Advanced wireless communication modules** used in portable and embedded systems requiring efficient and compact antenna solutions.

3.5. ADVANTAGES

- **Dual-band operation:** The podium-shaped geometry creates multiple effective current paths, enabling the antenna to operate at both 3.5 GHz and 5.3 GHz within a single compact structure.
- **Compact size:** The stepped design increases the effective current path length, allowing dual-frequency operation without increasing the overall antenna dimensions.
- **Improved impedance matching:** The microstrip line feed and stepped geometry help achieve better impedance matching, resulting in lower return loss and improved VSWR performance.
- **Low-cost fabrication:** The antenna is designed using an FR-4 substrate, which is inexpensive and widely available, making it suitable for mass production.
- **Simple and planar structure:** The design uses a single-layer microstrip configuration, enabling easy fabrication using standard PCB techniques.
- **Suitable for integration:** Due to its low profile and lightweight structure, the antenna can be easily integrated into modern wireless devices such as 5G modules and IoT systems.
- **Stable dual-band performance:** The proposed structure provides stable resonance and reliable radiation characteristics at both operating frequencies.

CHAPTER 4

RESULTS AND DISCUSSION

4.1. RESULTS

The proposed podium-shaped microstrip patch antenna was successfully designed and analyzed using CST Microwave Studio, and the simulation results confirm that the antenna operates effectively at the desired dual frequencies of 3.5 GHz and 5.3 GHz. The antenna achieved resonant frequencies at approximately 3.5 GHz and 5.36 GHz with return loss (S11) values of -17.9 dB and -22.1 dB, indicating excellent impedance matching between the feed line and the radiating patch. The simulated Voltage Standing Wave Ratio (VSWR) was found to be less than 2 at both frequencies, ensuring efficient radiation and minimal signal reflection. The results demonstrate that the incorporation of the podium-shaped geometry effectively modifies the surface current distribution, creating multiple current paths that enable dual-band operation. This validates the effectiveness of the proposed design for compact and efficient 5G and WLAN wireless communication applications.

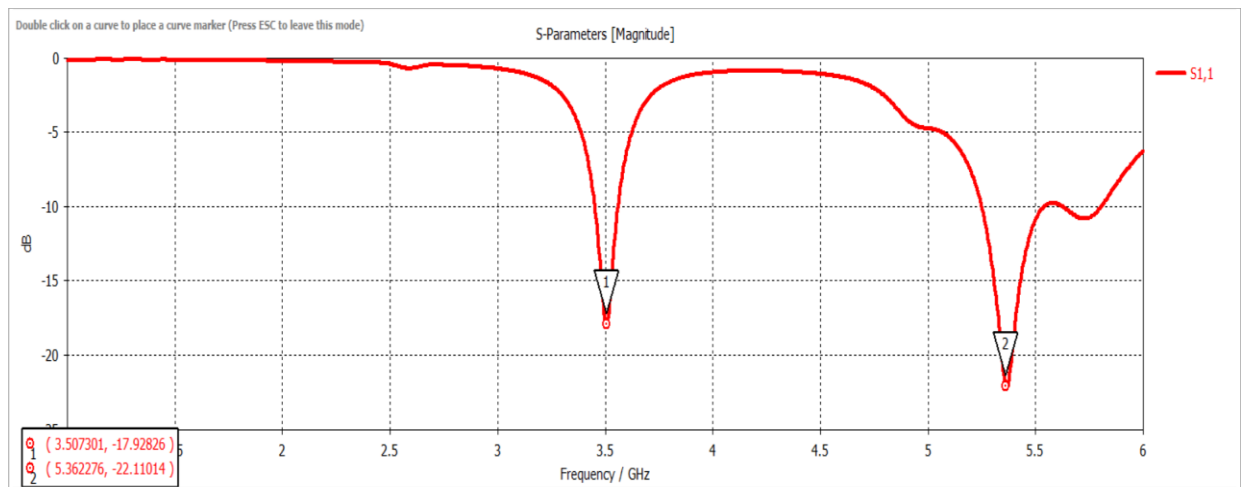


Fig.4.1 S-PARAMETER VALUE OF THE PODIUM-SHAPED MICROSTRIP PATCH ANTENNA

The S11 plot shows clear resonances at 3.5 GHz and 5.36 GHz with return loss values of approximately -17.9 dB and -22.1 dB, indicating good impedance matching. The deep notches confirm that most of the input power is radiated rather than reflected. This validates that the antenna operates effectively at the desired dual frequencies.

S.No	Frequency (GHz)	S11 (dB)
1	3.50	-17.93 dB
2	5.36	-22.11 dB

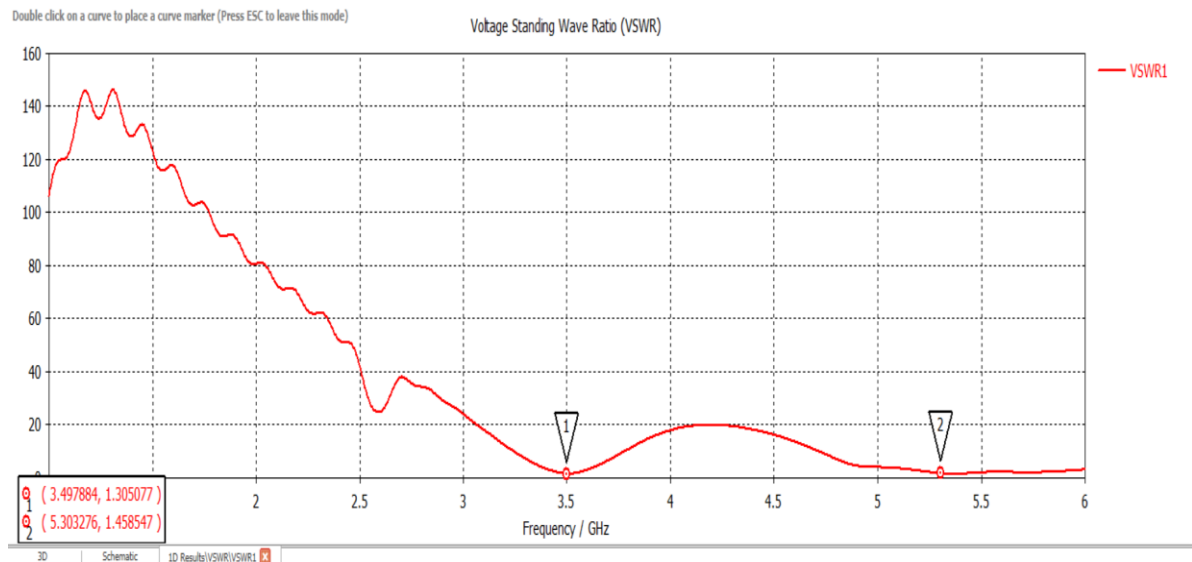


Fig.4.2 VSWR VALUE OF THE PODIUM-SHAPED MICROSTRIP PATCH ANTENNA

The VSWR plot shows minimum values close to 1 at approximately 3.5 GHz and 5.3 GHz, which are within the acceptable limit ($VSWR \leq 2$). This indicates efficient power transfer between the feed line and the antenna with minimal reflections. The results confirm stable and reliable antenna performance at both operating frequencies.

S.No	Frequency (GHz)	VSWR Value
1	3.50	1.30
2	5.30	1.46

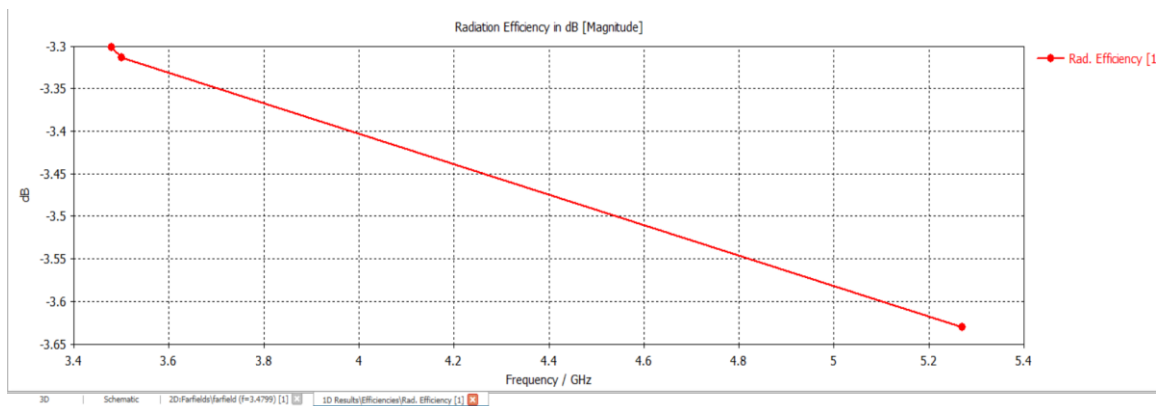


Fig.4.3 RADIATION EFFICIENCY OF THE PODIUM-SHAPED MICROSTRIP PATCH ANTENNA

Radiation efficiency decreases as frequency increases from 3.5 GHz to 5.3 GHz. It drops from about -3.3 dB to -3.63 dB, indicating higher losses at higher frequencies. Thus, the antenna is more efficient at lower frequencies.

S.No	Frequency (GHz)	Radiation Efficiency (dB)
1	3.50	-3.30 dB
2	5.30	-3.63 dB

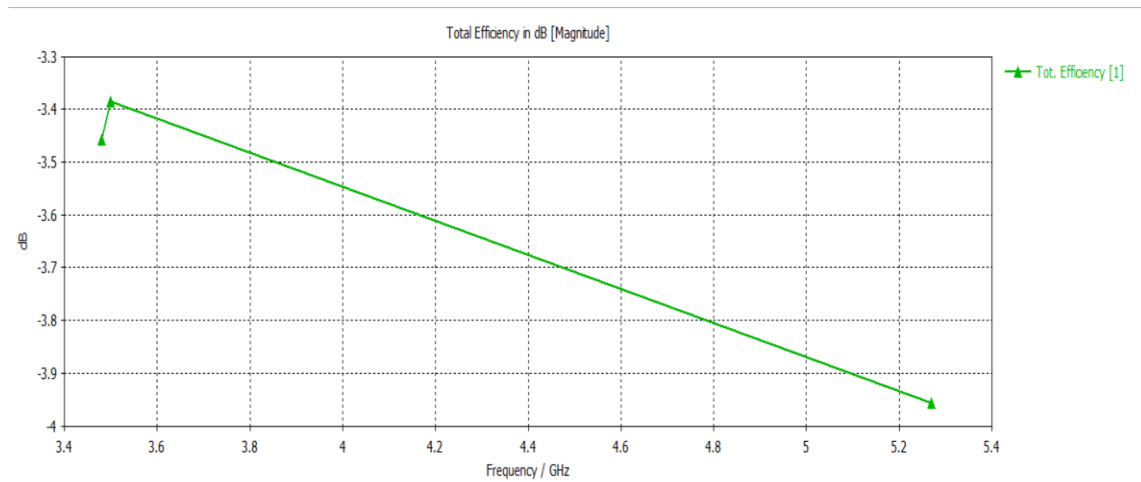


Fig.4.4 TOTAL EFFICIENCY OF THE PODIUM-SHAPED MICROSTRIP PATCH ANTENNA

S.No	Frequency (GHz)	Total Efficiency (dB)
1	3.50	-3.40 dB
2	5.30	-3.95 dB

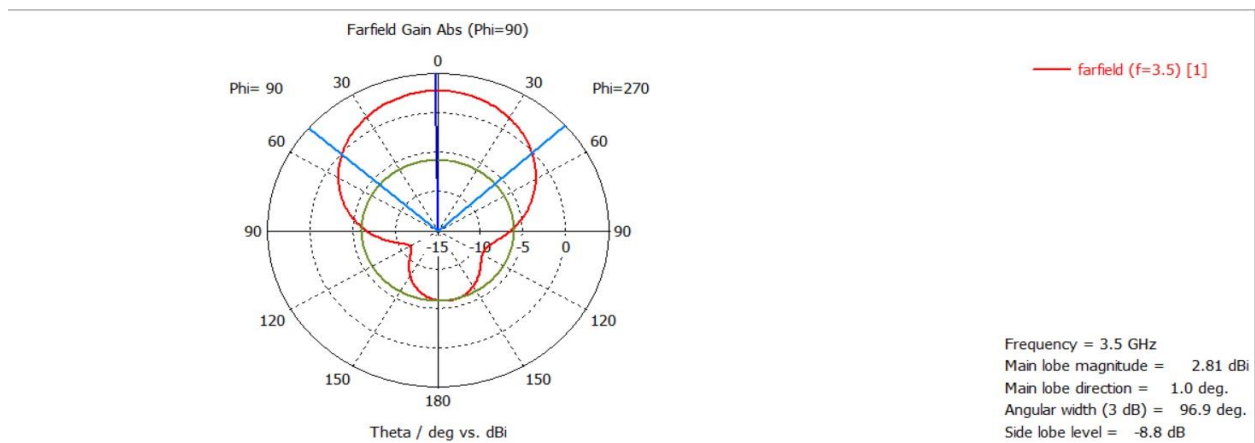


Fig.4.5 1D far-field radiation pattern of the antenna in 3.5 GHz

The 1D gain plot exhibits a directional radiation pattern with a peak gain of 2.81 dBi, indicating efficient signal transmission in the desired direction with improved directivity.

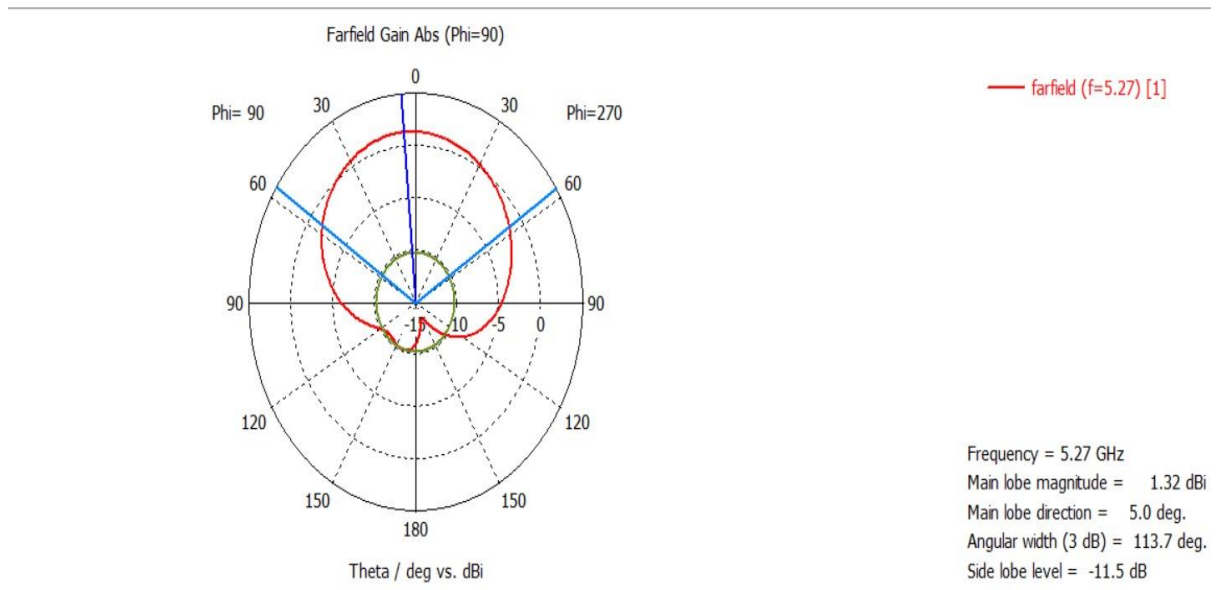


Fig.4.5 1D far-field radiation pattern of the antenna in 5.3 GHz

The 1D gain plot exhibits a directional radiation pattern with a peak gain of 1.32 dBi, indicating efficient signal transmission in the desired direction with moderate directivity.

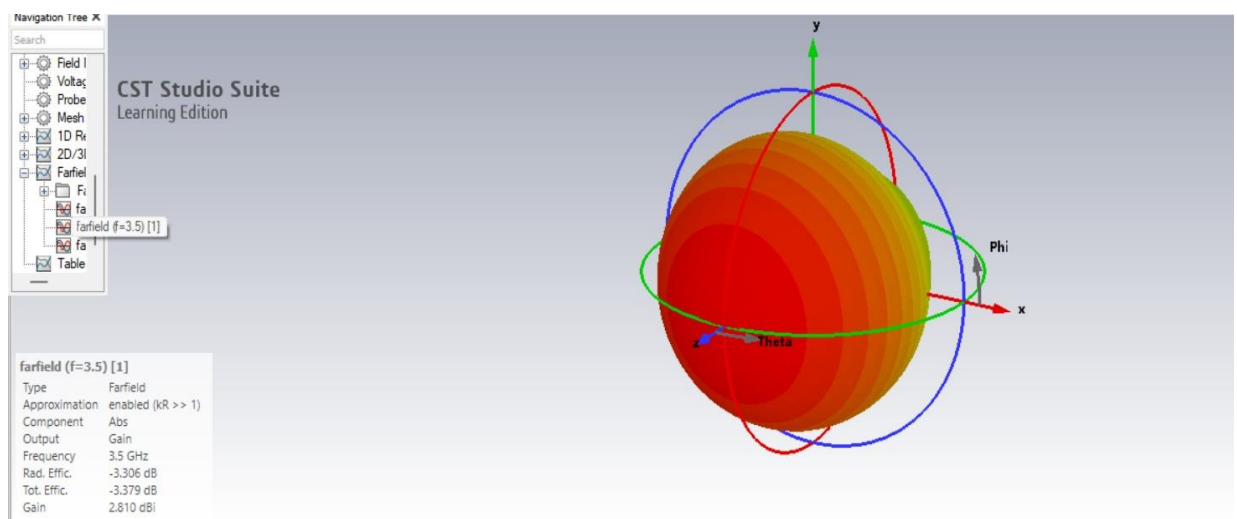


Fig.4.6 3D far-field radiation pattern of the antenna in 3.5 GHz

The 3D gain plot shows a peak gain of about 2.81 dBi at 3.5 GHz, with a directional radiation pattern and total efficiency around -3.38 dB, indicating stable antenna performance.

S.No	Frequency (GHz)	Gain (dBi)
1	3.50	~ 2.3 dBi
2	5.30	~ 2.0 dBi

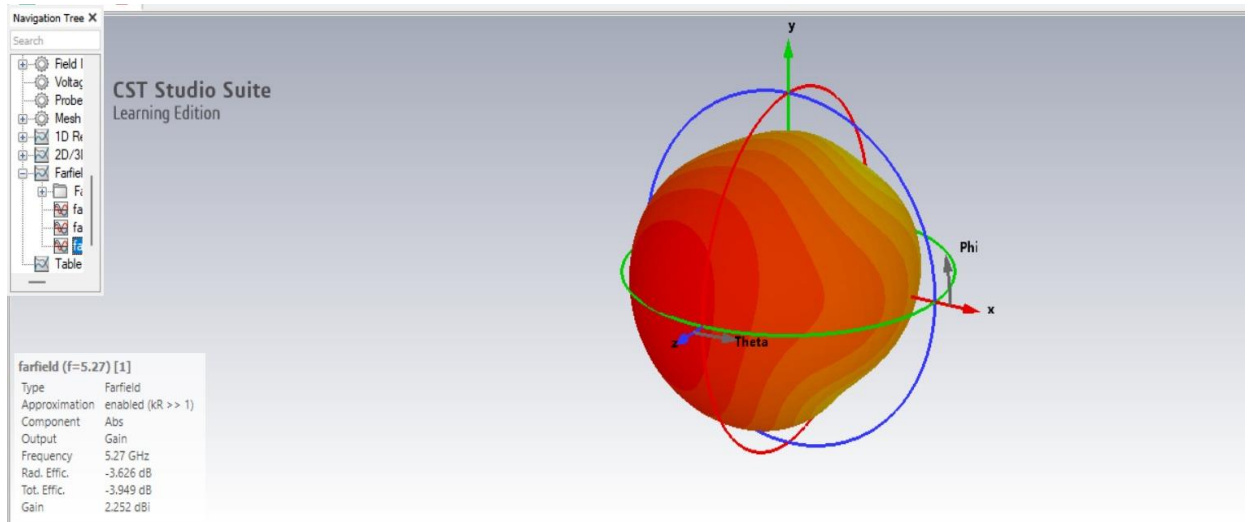


Fig.4.6 3D far-field radiation pattern of the antenna in 5.3 GHz

The 3D gain plot shows a peak gain of about 2.25 dBi at 5.27 GHz, with a directional radiation pattern and total efficiency around -3.95 dB, indicating stable antenna performance

S.No	Frequency (GHz)	Directivity (dBi)
1	3.50	~ 4.8 dBi
2	5.30	~ 4.5 dBi

4.2. DISCUSSION

This project demonstrates the design and analysis of a compact and efficient podium-shaped microstrip patch antenna suitable for modern wireless communication applications. The antenna is developed using an FR-4 substrate due to its low cost, easy availability, and suitability for practical implementation. Copper is used for both the patch and ground plane to ensure good conductivity and stable performance. One of the key strengths of this design is the use of a podium-shaped geometry, which provides better flexibility compared to conventional rectangular patches. This stepped structure modifies the surface current distribution, creating multiple effective current paths that enable dual-band operation. The antenna is designed to operate at 3.5 GHz and 5.3 GHz, making it suitable for Sub-6 GHz 5G (n78 band) and WLAN applications.

The performance of the antenna is evaluated using simulation tools, where important parameters such as return loss (S_{11}) and Voltage Standing Wave Ratio (VSWR) are analyzed. The observed results indicate strong resonance at both operating frequencies, with return loss values below -10 dB and VSWR less than 2, confirming efficient signal transmission and

minimal reflection. The radiation pattern is stable and nearly omnidirectional, ensuring reliable communication performance. The compact design of the antenna makes it suitable for integration into modern wireless devices. Additionally, the structure is simple and easy to fabricate using standard PCB techniques. Overall, this project highlights effective antenna design, dual-band performance optimization, and suitability for advanced 5G and WLAN wireless systems, making it valuable for both academic research and practical applications.

Parameter	3.5 GHz Value	5.3 GHz Value	Inference
Return Loss (S11)	-17.93 dB	-22.11 dB	Good matching
VSWR	1.30	1.46	Efficient transmission
Gain	~ 2.3 dBi	~ 2.0 dBi	Moderate gain
Directivity	~ 4.8 dBi	~ 4.5 dBi	Moderate directivity
Total Efficiency	~ -3.40 dB (~45%)	~ -3.95 dB (~40%)	Moderate efficiency
Radiation Efficiency	~ -3.30 dB	~ -3.63 dB	Acceptable radiation
Radiation Pattern	Omnidirectional	Omnidirectional	Stable radiation

CHAPTER 5

CONCLUSION & FUTURE SCOPE

5.1. CONCLUSION

In this project, a novel podium-shaped microstrip patch antenna was designed and simulated to operate at dual frequencies of 3.5 GHz and 5.3 GHz for modern wireless communication applications. By incorporating a stepped (podium) geometry into the patch, the proposed design successfully modifies the surface current distribution, enabling dual-band operation while maintaining a compact structure. The antenna, designed on an FR-4 substrate and analyzed using CST Microwave Studio, achieved return loss values of -17.9 dB and -22.1 dB, along with $VSWR \leq 2$ at both frequencies, demonstrating efficient impedance matching and reliable performance. The results confirm that the podium-shaped geometry provides an effective alternative to conventional patch designs, offering a simple, low-cost, and easily integrable antenna solution for 5G and WLAN applications.

Reference	Frequency (GHz)	S11 (dB)	VSWR	Patch Shape
[3]	3.5	-12.5	1.75	Rectangular
[5]	5.3	-14.2	1.68	U-slot
[7]	3.5 / 5.3	-10.8	1.92	Modified Rectangular
This Work	3.5 / 5.3	$-17.9 / -22.1$	≤ 2	Podium-Shaped

TABLE 5.1. Comparison of the Proposed Antenna with Reference Designs at 3.5 GHz and 5.3 GHz

The table compares the proposed antenna with existing designs at the 3.5 GHz and 5.3 GHz bands. It is observed that the proposed antenna achieves better return loss and acceptable VSWR compared to conventional rectangular and modified designs. Thus, the proposed podium-shaped antenna provides improved impedance matching and efficient dual-band performance.

5.2. FUTURE SCOPE

The proposed podium-shaped microstrip patch antenna can be further improved and extended in several ways to enhance its performance and broaden its range of applications. In future work, the antenna can be fabricated and experimentally tested using standard measurement setups such as an anechoic chamber to validate the simulated results. The design

can also be modified to support multiband or wideband operation by optimizing the stepped geometry and current paths, enabling operation at additional frequencies.

The antenna performance can be further enhanced by improving gain and efficiency using array configurations or low-loss substrate materials. Additionally, the design can be adapted for circular polarization and integrated into compact wireless systems to support advanced 5G and WLAN communication applications.

5.3. REFERENCES

1. C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed., Wiley, 2016.
2. K. L. Wong, *Compact and Broadband Microstrip Antennas*, Wiley, 2002.
3. D. M. Pozar, “Microstrip Antennas,” *Proceedings of the IEEE*, vol. 80, no. 1, pp. 79–91, 1992.
4. R. Garg, P. Bhartia, I. Bahl, and A. Ittipiboon, *Microstrip Antenna Design Handbook*, Artech House, 2001.
5. J. R. James and P. S. Hall, *Handbook of Microstrip Antennas*, IET, 1989.
6. H. Wong, K. M. Luk, and C. H. Chan, “Microstrip Patch Antennas for Wireless Applications,” *IEEE Access*, vol. 5, pp. 254–267, 2017.
7. CST Studio Suite, Dassault Systèmes, “CST Microwave Studio User Manual,” 2023.
8. T. Huynh and K. F. Lee, “Single-Layer Single-Patch Wideband Microstrip Antenna,” *Electronics Letters*, vol. 31, no. 16, pp. 1310–1312, 1995.
9. R. Chair, C. L. Mak, K. F. Lee, K. M. Luk, and A. A. Kishk, “Miniature Wide-Band Half U-Slot and Half E-Shaped Patch Antennas,” *IEEE Transactions on Antennas and Propagation*, vol. 53, no. 8, pp. 2645–2652, 2005.
10. S. Maci and G. B. Gentili, “Dual-Frequency Patch Antennas,” *IEEE Antennas and Propagation Magazine*, vol. 39, no. 6, pp. 13–20, 1997.